

<EE-2003>- <Computer Organization and Assembly Language>

Form number	COURSE OUTLINE / DOCUMENT	
COURSE INSTRUCTOR INFORMATION	Name	Khalid Hussain
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DEGREE INFORMATION	Program	Batch	Section(s)				Semester	FALL
	BSCS	22	C	D	E		Year	2023

COURSE INFORMATION	Course Category C- Core/ E-Elective		Code	Title	Credit hours
	C		<u>EE-2003</u>	<u>COAL</u>	3
	Prerequisite(s)		EE-109	<u>DLD</u>	4
	TA Required (Yes/ No)	No. of TA(s)	Brief Justification		

TEXT BOOK(S) INFORMATION	Title of Book	Assembly Language for x86 Processors	Edition 6th
	Author(s)	Kip R. Irvine	
	Publisher	Pearson	

TEXT BOOK(S) INFORMATION	Title of Book	The Elements of Computing Systems	Edition 2nd
	Author(s)	Noam Nisan & Shimon Schocken	
	Publisher	MIT Press	

Reference Book (s)	1.	Title of Book	Assembly Language Programming
		Authors	Belal Hashmi, Junaid Haroon
	2.	Title of Book	x86-64 Assembly Language Programming with Ubuntu
		Imprint details	Ed Jorgensen
	3.	Title of Book	
		Imprint details	

Short Description	This course provides an overview of the architecture and organization of a computer, such as the CPU, memory, I/O organization, peripherals and so on. From this course, students will learn the basics of computer architecture and low level programming. i.e. assembly code and hardware manipulation.
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Description of Course: (not more than 250 words)	This course will cover the fundamentals of computer organization and machine instructions, as well as memory access and storage, instruction execution, assembly language, computer organization, data representation and transfer, digital arithmetic, memory storage and addressing methods, procedures and interrupts, conditional processing, and other topics. Assembly language is the basis for current computer software. You will also learn about the components of a normal computer system and how the operating system controls these components while you study assembly language. Students who successfully complete this course will have a comprehensive grasp of computer organization and design, as well as the ability to evaluate and execute various real issues using Assembly language.
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Course Objectives (CO): (Brief & unambiguous)... at least 5 COs

- a. tend to describe specific, discrete units of knowledge and skill
- b. can be accomplished within a short time frame - still may be relevant for a class period
- c. tend to be **STATEMENTS OF INTENT**; do not necessarily suggest that the behavior has been demonstrate

1.	To introduce basic concepts of computer organization with emphasis on the lower level abstraction of a computer system including instruction set architecture, addressing modes, memory models and assembly language programming.
2.	To illustrate the computer organization concepts by Assembly Language programming
3.	Introduction to x86 Architecture.
4.	Familiarization with Assembly Language directives, macros, operators, and program structures.
5.	Learning Programming methodology to be able to create system level software tools and application programs
6.	Understanding of interrelationship between hardware and software
7.	Comparison between different processors families

Learning Outcome (LO): (Brief & unambiguous-with reference to course objectives i.e. at least 5 LOs

- a. describe broad aspects of behavior which incorporate a wide range of knowledge and skill
- b. accomplished over time in several learning experiences
- c. refer to **DEMONSTRATIONS OF PERFORMANCE**

a.	Understanding of the major components of a computer and basics of computer organization
b.	Understanding of interconnection between computer modules and functions
c.	Understanding of the internal working of central processing unit
d.	Understanding of Instruction set architecture. addressing modes and formats
e.	Understanding of Assembly language Programming Concepts for intel x86 processor (Data types, Functions Control Structures, Loops, Procedures, Arrays, Pointers, Stack, Strings, Interrupts, file handling)
f.	Understanding of direct Hardware interaction using assembly language
g.	Graphical user interface development and controlling GUI through keyboard and mouse

Courseware Structure: (Mark X where applies)

Lecture (Lect)	Multimedia (MM)	Exercise (s) (Exer)	Labs (Lab)	Case Studies (CAS)	Assignment (s) (Assign)	Group Presentation (G-Pres)	Any other Medium Midterms
X	X	X	X	X	X		X

COURSE CONTENTS:

Weeks	Contents/Topics	** Courseware Events (MM/ IT Lab/Case Study/ Assignment/ Presentation etc.)	Comments (if any)
Week-01	Introduction to the elements of computing systems Review of the combinational logic design (Logic gates and their abstractions, Flipflops and their usage in designing registers, ALU, and RAM)		

Week-02	<u>Abstraction to Instruction</u> Introduction to Assembly Language Types of Architectures, Intel Microprocessors, Understanding the first program (Assembler, Linker, and Debugger) Memory models		
Week-03	<u>Addressing Modes</u> Data declaration and directives Direct addressing Register indirect addressing Register + offset addressing Other types of addressing modes		
Week-04	<u>Branching</u> Conditions Jumps (Types of Jumps) Conditional Jumps		
Week-05	<u>Teching Multiplication to the computer</u> Shifting and rotations Multiplication in assembly Extended multiplication Logical and bitwise operations		
Week-06	<u>Revision</u>		
Week-07	<u>Stack</u> Intro to stack and subroutines Stack operations		
Week-08	<u>Accessing the display memory</u> Display memory and ASCII codes Finding the location in the display memory DIV and MUL instructions		
Week-09	<u>String Processing</u> String Instructions with examples		
Week-10	<u>Interrupts</u>		
Week-11	<u>Software Interrupts</u> Bios and DOS interrupt with examples		
Week -12	<u>Revision</u>		
Week 13	<u>Real-time Interrupts</u> I/O ports		
Week-14	<u>Computer Arithmetic</u> Signed Multiplication		
Week-15	<u>Computer Arithmetic</u> Fast Multiplication Integer Division Floating Point Numbers		
Week-16	<u>Revision & System on chip</u>		

**** Courseware Events** column is subject to variations / appropriations

Grading Criteria		
√	ABSOLUTE Grading	RELATIVE Grading

Marks Distribution:

Particulars	% Marks	*Weight Ranges
1. Assignments	10	
2. Quizzes	20	
3. Mid Term-I	10	
4. Mid Term-II	15	
6. Final Exam	40	
7. Any other-Class performance	5	
Total:-	100	100

*Weight Ranges as defined at NeON

Planned Courseware Events:

Particulars	Planned (Qz/As/Labs)	Remarks
1. Quizzes	6	All quizzes announced
2. Assignments/	>2 & <6	
3. Presentations		
4. Labs	< or = 14	Individual Lab Exercises
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Academic and Moral Integrity:

All assignments should be your own work (or your group's when approved). PLAGIARISM will be awarded with "F" grade and/or reported to the University for academic and moral misconduct. To protect yourself, ALWAYS PROVIDE REFERENCES!

SEPECIAL INSTRUCTIONS (Specific to the subject being taught)

Please try to do your own work/assignments honestly. In this way we will be able to identify the students who need more help in studies.

Your textbooks should be the main source of all your work. All of you are highly recommended to read the textbook in order to get good knowledge of Assembly Language.